

Monitoring Glacier Retreat in Switzerland through Attention U-Net Segmentation

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MSc Artificial Intelligence for Sustainable Development (2025-2026)



Deforestation Detection from Satellite Imagery

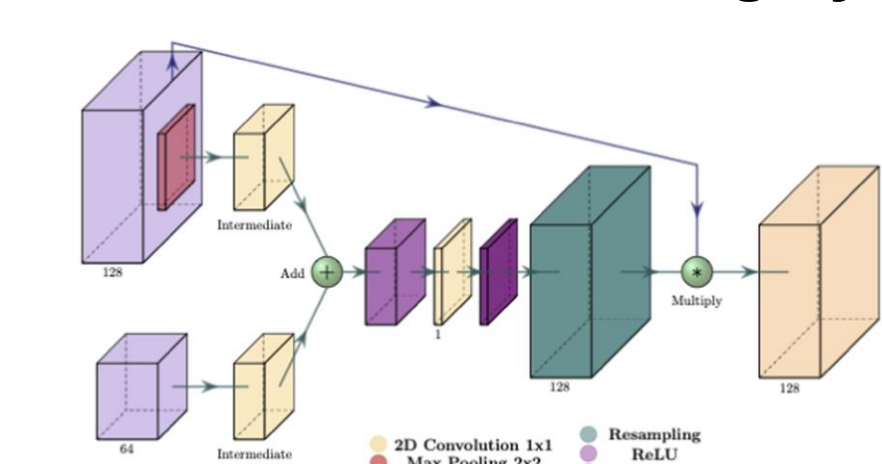
Amazon Rainforest covers about **40% of Earth's tropical forests**, and serves as a habitat for roughly **10% of all species** worldwide

Large-scale deforestation
(about 2300 hectares/day (2020))

A major **threat** to species, a **contributor** to accelerating climate impacts, and a **danger** to the forest's long-term survival

Attention U-Net Model to monitor deforestation from Sentinel-2 satellite imagery

- Encoder: 4 down sampling blocks
- Decoder: 4 up sampling blocks
- Output: Sigmoid activation
- Loss: Binary Cross-Entropy
- 4-band Amazon Forest data model



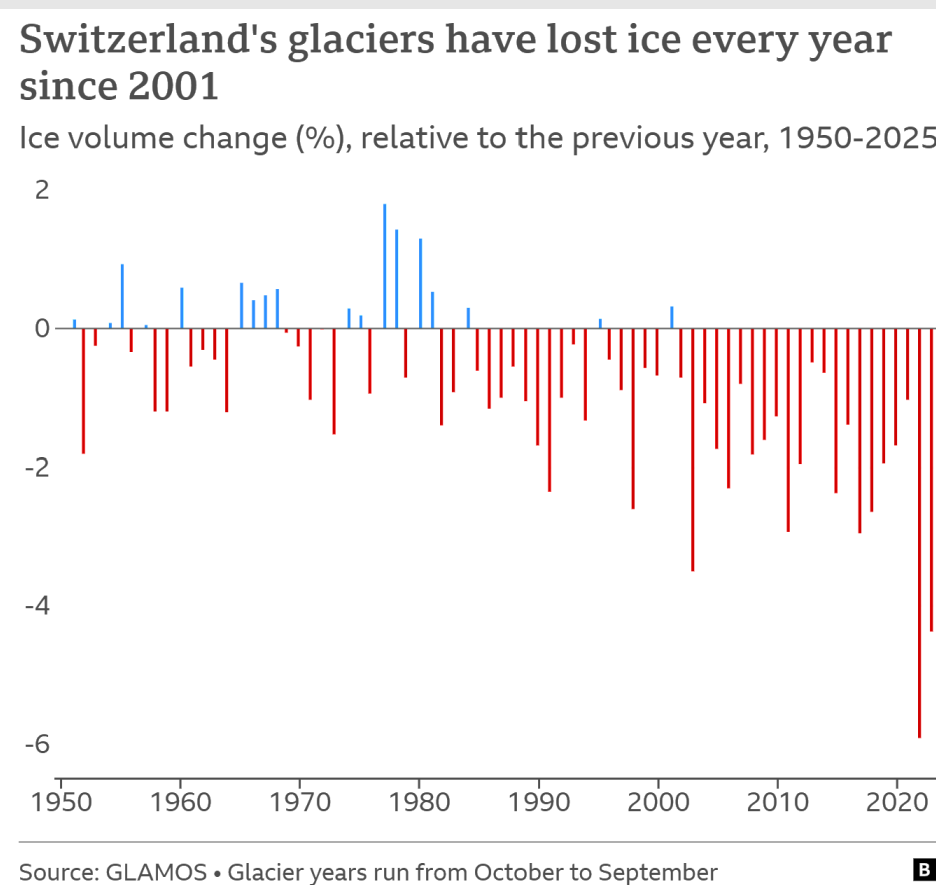
Strong performance:

- Validation set (own replication): F1-Score = 0.9239
- Test set (own replication): F1-Score = 0.8940

Monitoring Glacier Retreat in Switzerland

Because of global warming, **Switzerland's glaciers are melting at an alarming rate:**

- Swiss Alps have **lost 25%** of its volume since 2015
- **Over 1k** small glaciers have already disappeared
- **10% decrease in volume** between 2022 and 2023
- One of **Europe's largest freshwater reservoir**



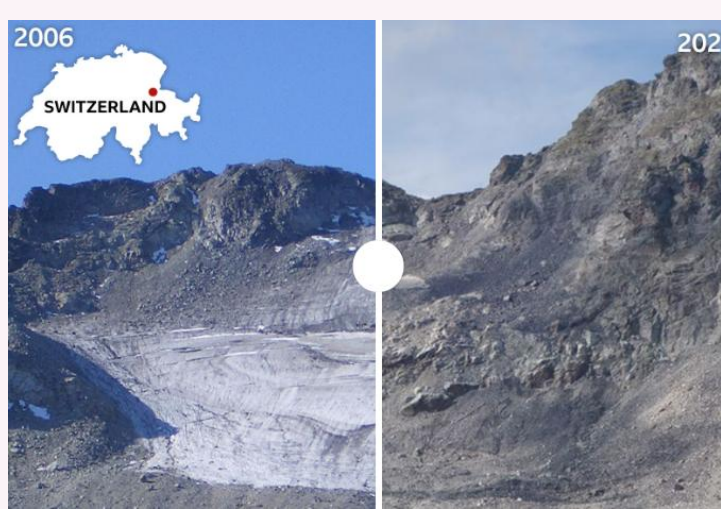
Unstable **water supply**

Hydropower instability

Increase in **natural hazards**

Impact on **biodiversity**

Tourism, cultural value and global heritage



Monitoring Glacier Retreat:



Early detection of collapses and unstable slopes → strengthen resilience to climate-related hazards (13.1)
Continuous glacier data for hydropower planning → integrate climate measures into national policies (13.2)
If transparent monitoring → improved climate **awareness** (13.3)

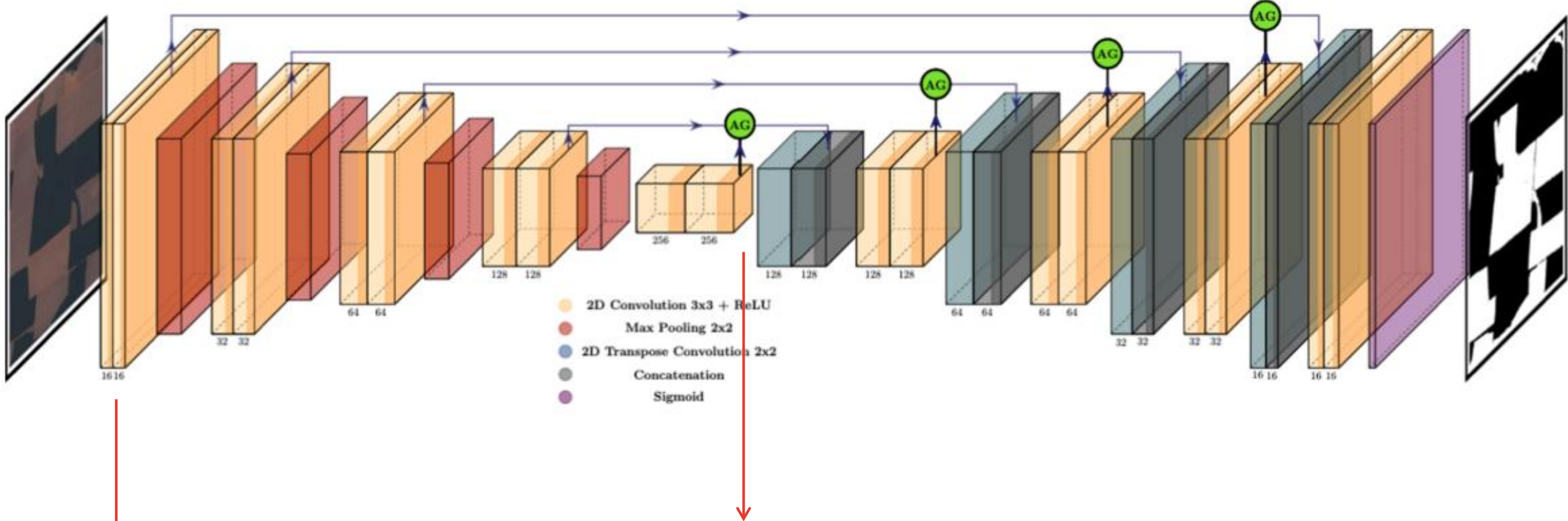
Observations inform LT freshwater availability → **increase water-use efficiency** (6.4)
Safeguard hydrological systems (6.6)



Supports **early-warning** for communities downstream → reducing disaster risks and impacts (11.5)



Adapted Model Architecture



Add dropout layer (50% dropout rate)

Retrained from scratch to adapt to new input size
(512,512, 4) → (512,512, 6)

Loaded pre-trained Amazon weights → full fine-tuning of all layers

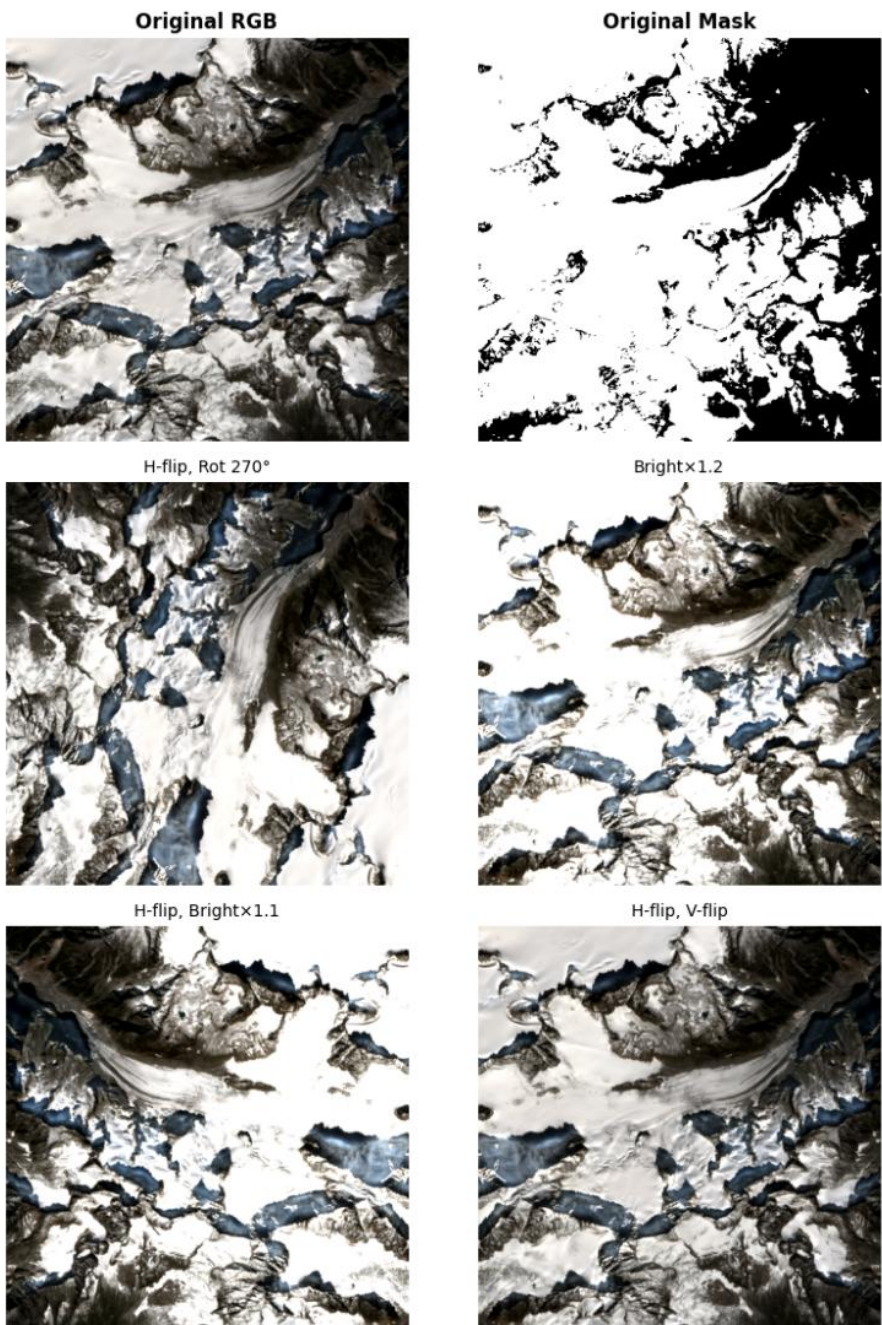
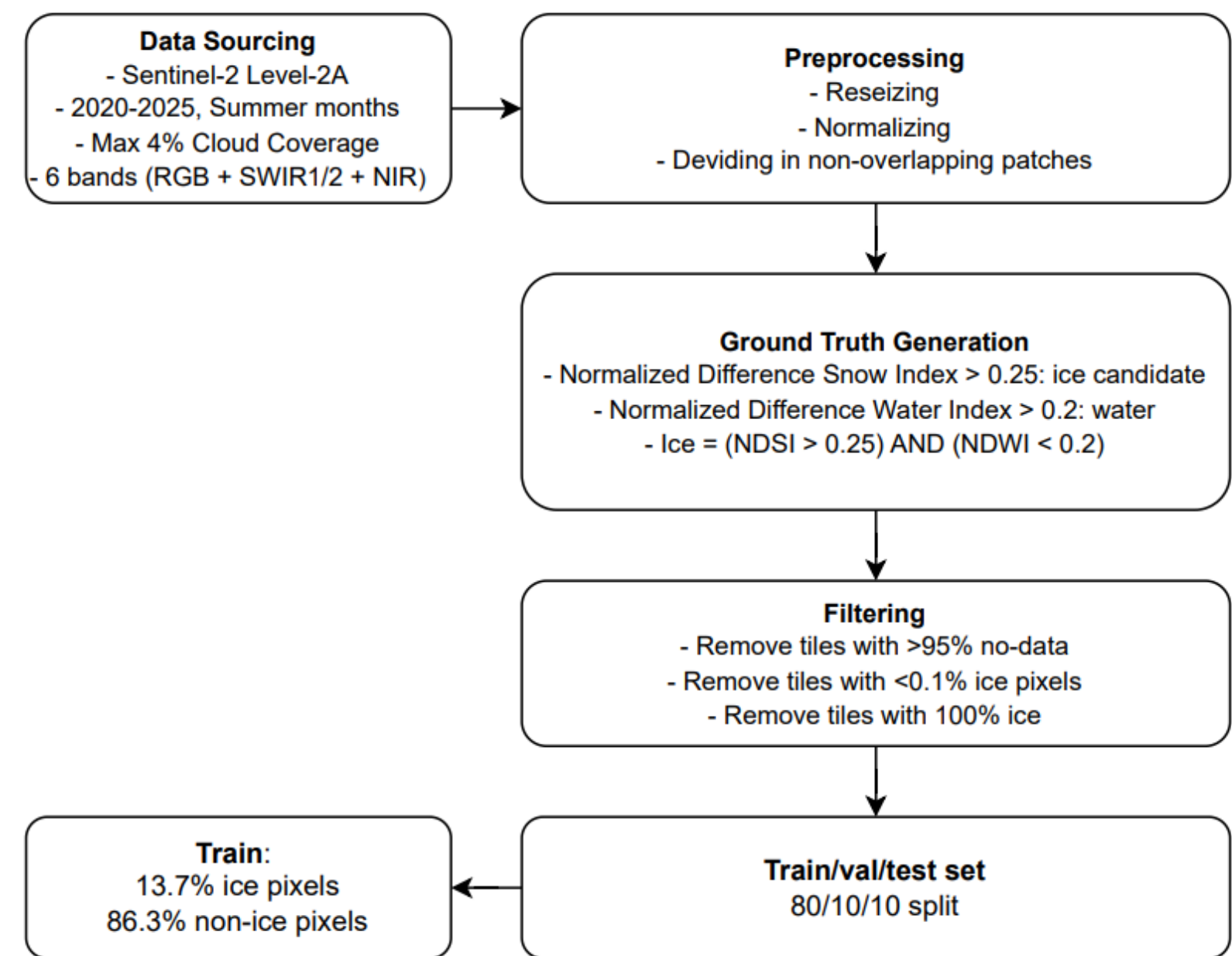
Other adaptations:

- Learning rate scheduler
- SGD → Dice Loss
- Data Augmentation
- Batch Size
- Early Stopping

Optimal parameters:

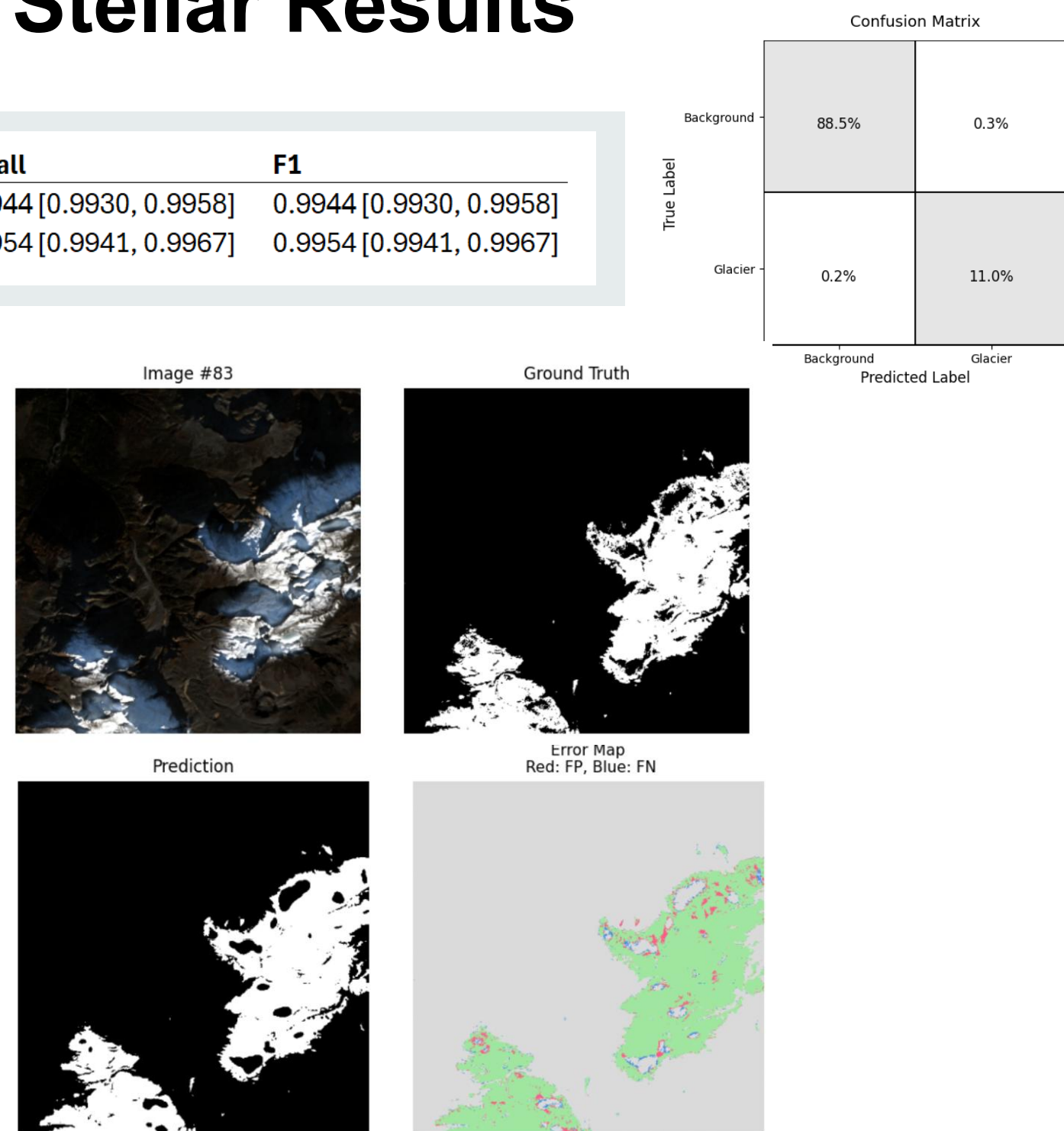
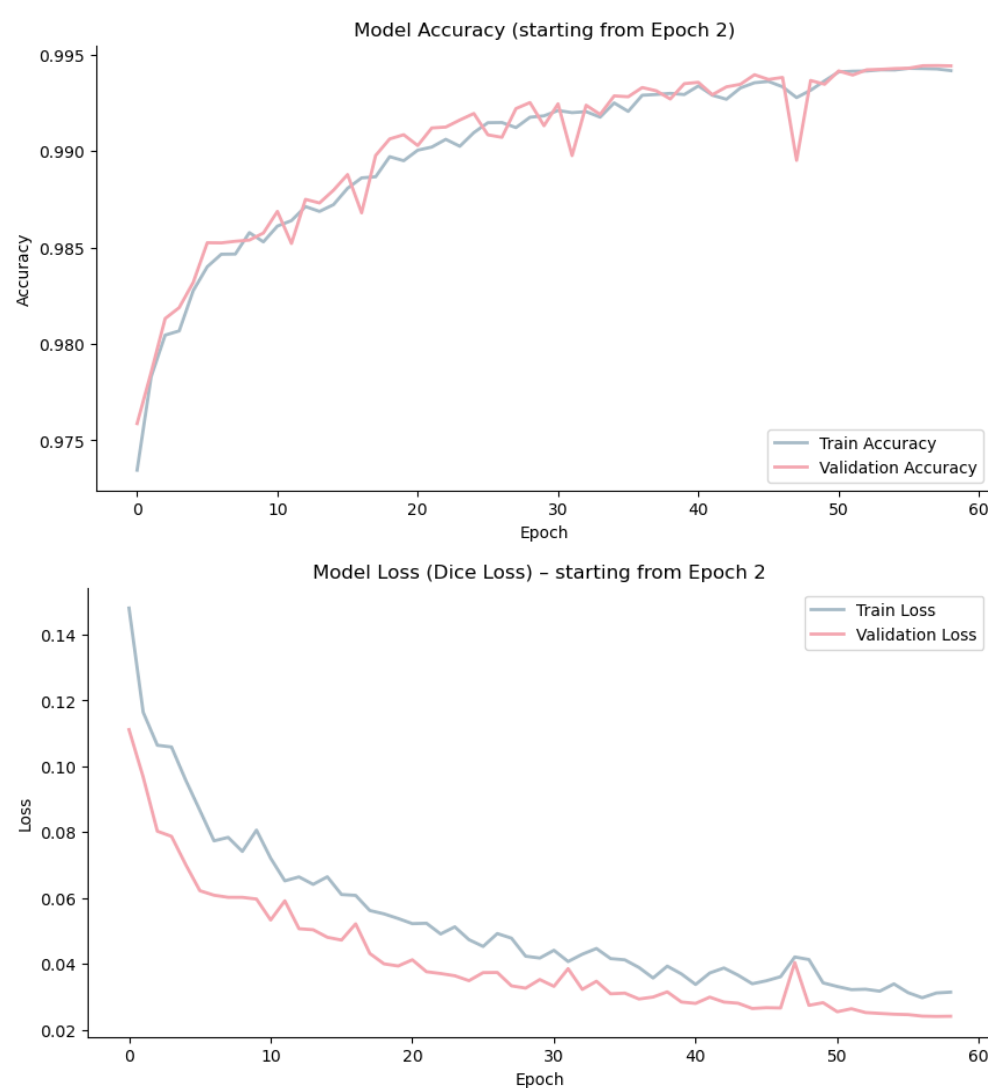
```
PARAMETERS = {
  'batch_size': 2,
  'epochs': 60,
  'loss': Dice Loss,
  'learning_rate': 0.0001,
  'reduce_learning_rate_factor': 0.3,
  'patience_learning_rate': 5,
  'drop_rate': 0.5,
  'patience_early_stopping': 10}
```

Sentinel-2 satellite images of Swiss Alps (Jungfrau-Aletsch region)

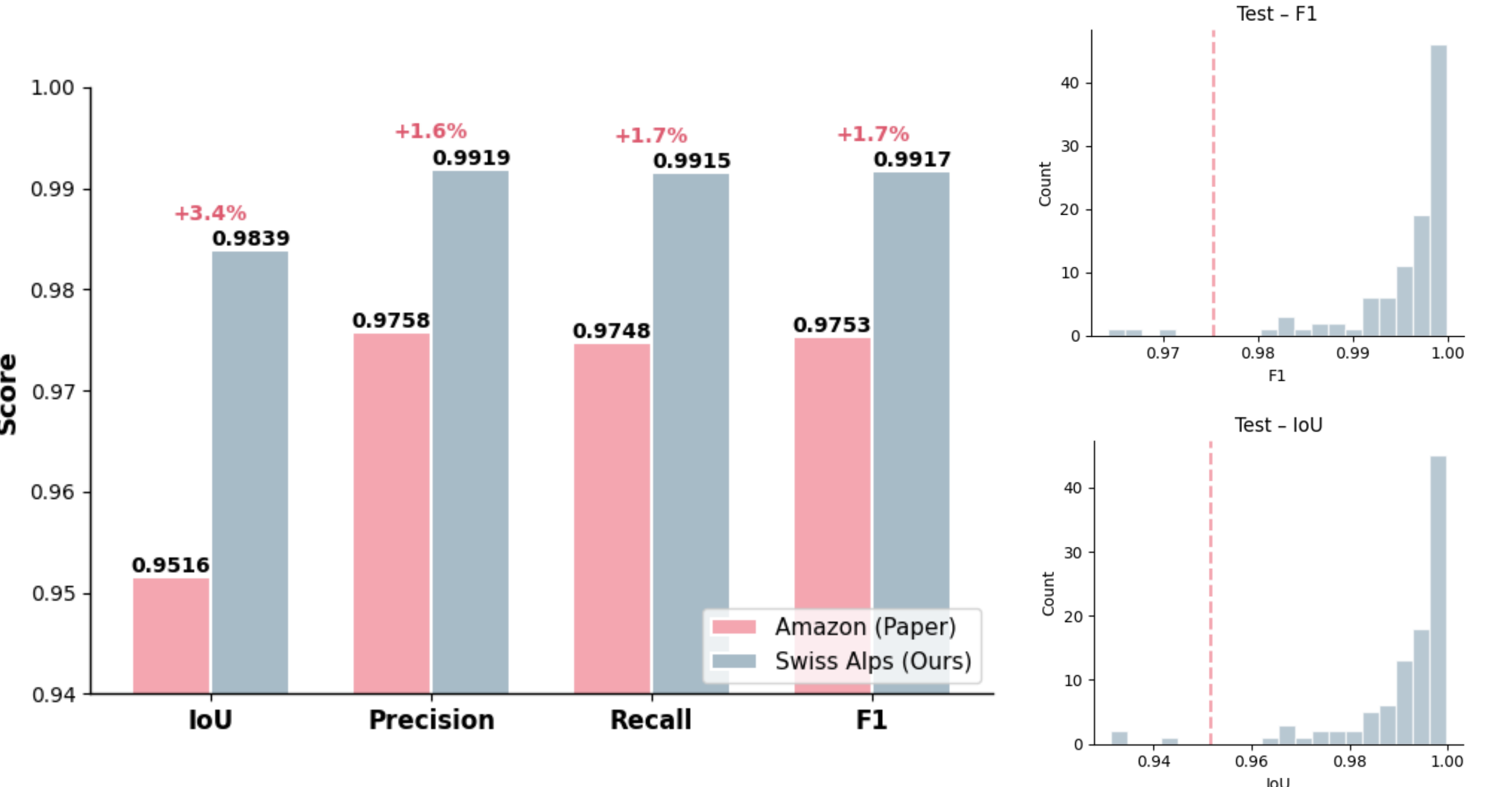


Adapted Model Achieves Stellar Results

Split	IoU	Precision	Recall	F1
Validation	0.9892 [0.9865, 0.9919]	0.9945 [0.9931, 0.9959]	0.9944 [0.9930, 0.9958]	0.9944 [0.9930, 0.9958]
Test	0.9910 [0.9885, 0.9936]	0.9954 [0.9942, 0.9967]	0.9954 [0.9941, 0.9967]	0.9954 [0.9941, 0.9967]



Difference in Performance is Statistically Significant



Discussion and Critical Reflections on SDGs

Why Better Performance?

- Task characteristics: Ice has unique spectral signature
- More spectral info: 6 bands vs 4 bands
- More training data (800 vs. 200) + augmentation
- More advanced model (dropout, LR scheduler)

Sustainability Trade-offs:

- Trained on one region → generalisable?
- No community engagement, who is in control?
- Impact depends on whether **policymakers** and local authorities **act on the generated insights**
- **Communities** downstream need to receive timely accessible information
- **Benefit:** Scalable, frequent monitoring
- **Cost:** Infrastructure dependencies (satellites, GPUs), carbon cost for training